



LUMEN

A Task-Oriented Assistive System for Blind People

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MOTIVATION

Indoors, blindness still means dependence

2.2 B

people live with visual impairment
worldwide

39 M

are completely blind

\$15–50 K

cost of a trained guide dog

Why indoors is the hard case

- GPS is unavailable; layouts are unfamiliar and unmapped
- The white cane senses only within arm's reach — no advance information
- Employment, social participation, and independence all hinge on unsolved indoor mobility

Existing systems pick the wrong two extremes



Continuous narration

- Describes everything, all the time
- User must filter a constant audio stream while listening to the world

→ *cognitive overload*



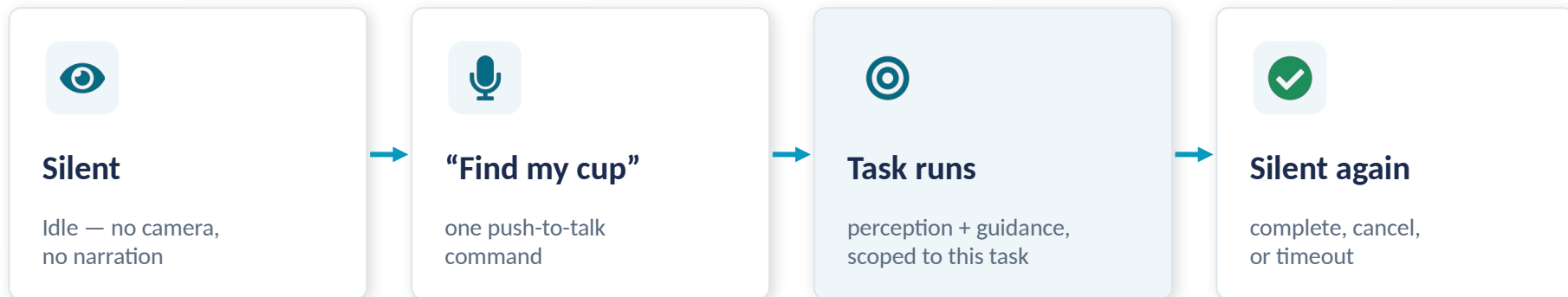
Query-based assistance

- Answers only what is asked
- In an unfamiliar space, the user cannot know what to ask, or when

→ *burden shifts to the user*

The missing middle ground: perception scoped to explicit tasks — active during them, structured by them, silent outside them.

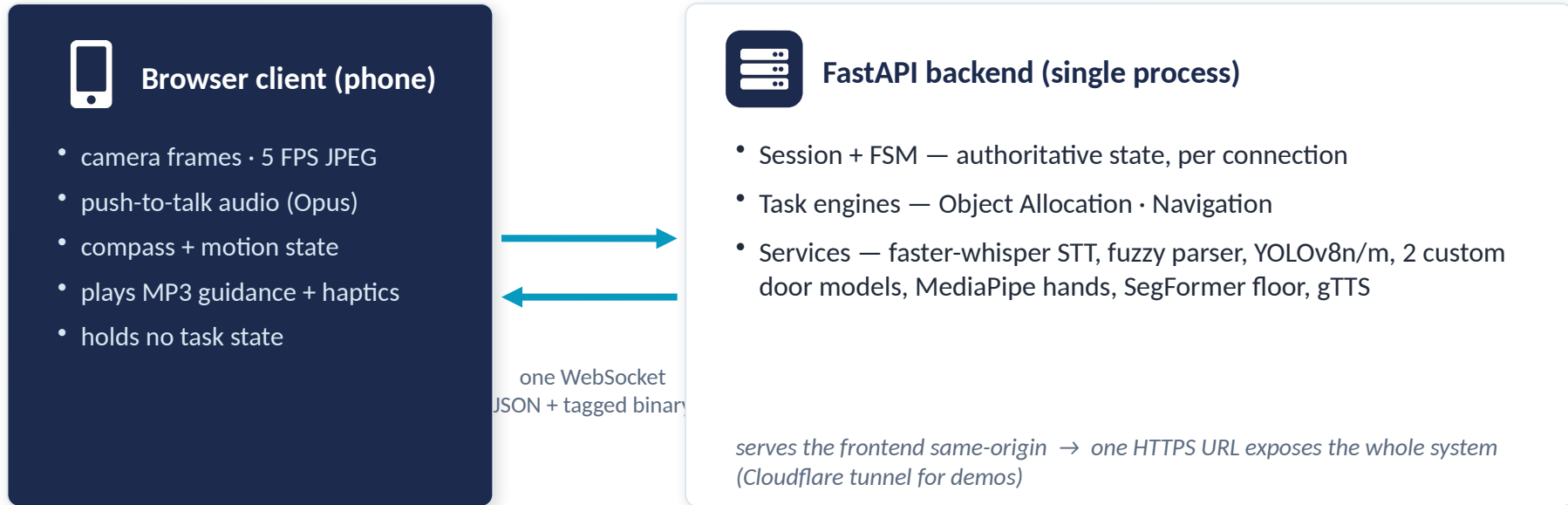
Task-scoped perception, delivered with zero install



No dedicated hardware. No app. No install.

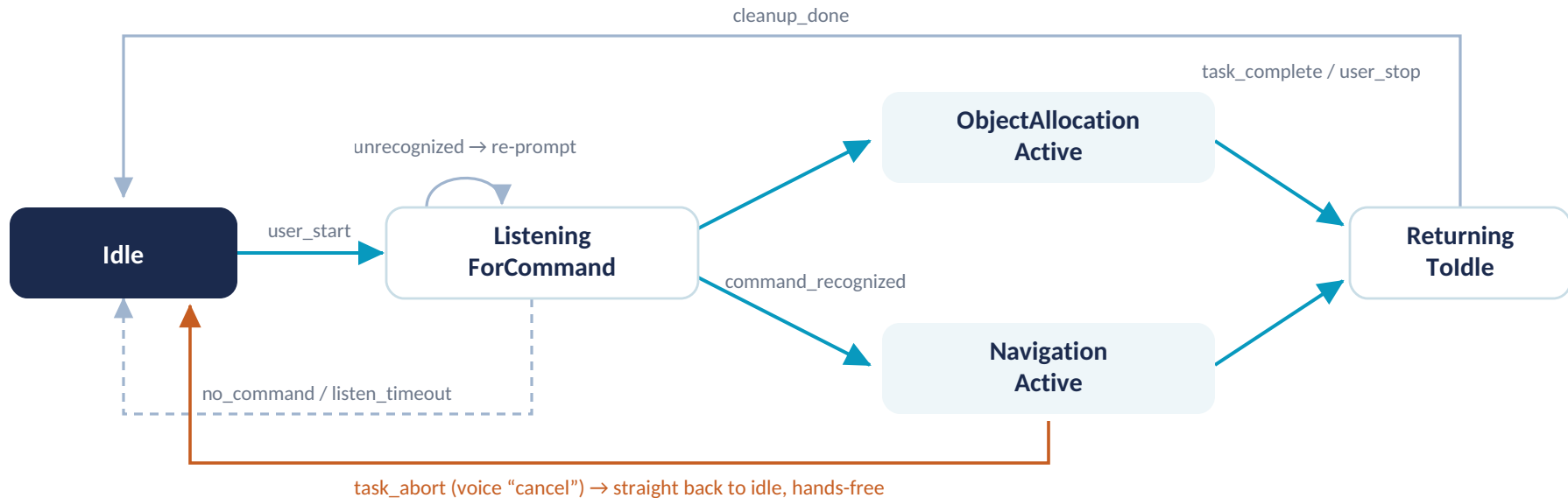
- The user opens a web page on an ordinary smartphone — camera, mic, speaker, compass already in their pocket
- Two task families: Object Allocation (find it, reach it, grasp it) and Navigation (walk me to the kitchen)

Thin client, smart server, one WebSocket



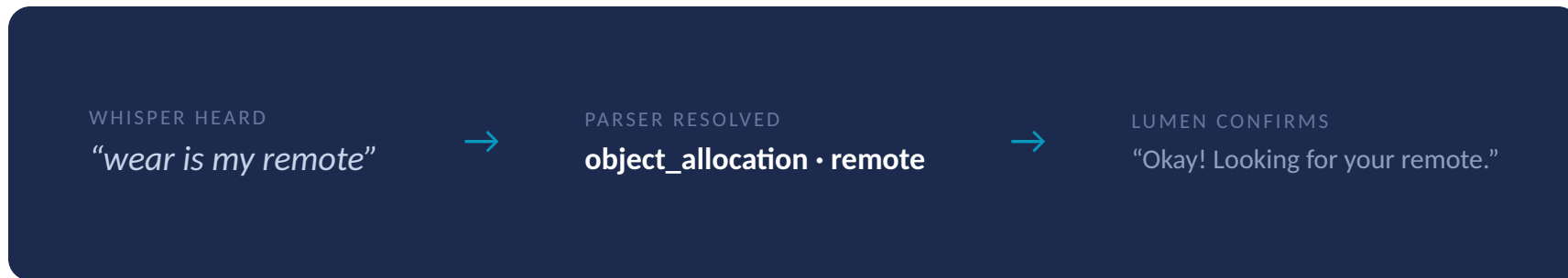
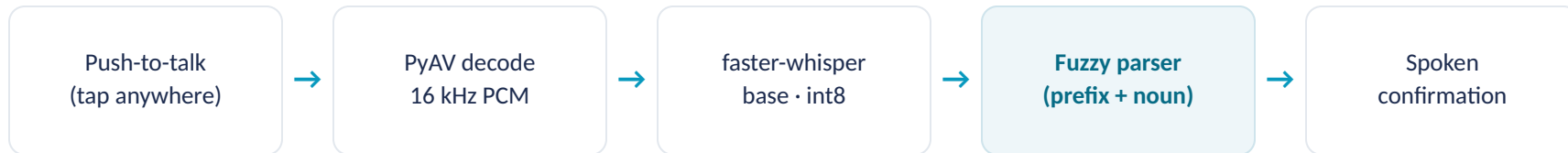
Skips frames, never lags: task loops always read the latest frame; all inference runs in worker threads.

A finite state machine governs every session



- Every transition is an explicit event — behavior is predictable, interpretable, and testable
 - Entry/exit hooks start and cancel task loops: a task can never outlive its state
- Silence is enforced by structure, not discipline: no state exists in which the system can speak outside a task

From a press of the screen to a confirmed task



→ For a blind user, the confirmation is the interface: nothing runs until Lumen says back what it understood

Object Allocation — “Find my cup”

SEARCHING

YOLOv8n at 5 FPS,
restricted to the
requested class —
cheaper, and nothing else
can distract it

BEHAVIOR

scan prompt every 8 s
— never nags faster

SIGHTED

Detectors blink: an object
can appear in one frame
and vanish in the next

Lumen announces the
cup only once it holds in
3 of the last 5 frames

LUMEN

“Found your cup, to
your left”

GUIDING

Distance = how much of
the frame the object fills
— “near” is 40% for a
laptop, 18% for a cup

Direction = left / ahead /
right — a change must
hold 2 frames; user
movement resets it

BEHAVIOR

speaks on real change
only • ≤ every 6 s

LOST

One-shot memory of
where it was; **60 s** with
no sighting → give up and
apologize

LUMEN

“It was to your left”

DONE

Two endings: found it
without **hand guidance**
→ the user confirms

Under hand guidance,
the touch is the proof —
Lumen completes the
task itself

USER

“Got it”

LUMEN • ON TOUCH

“Grasp it”

→ Even inside a task, silence is the default: Lumen speaks only what has earned trust, and only when it changes

Reach Guidance — steering the hand, not the body

approach

“Move your hand to the right”

one cue at a time, along the dominant axis



almost

“Almost there. Reach forward”

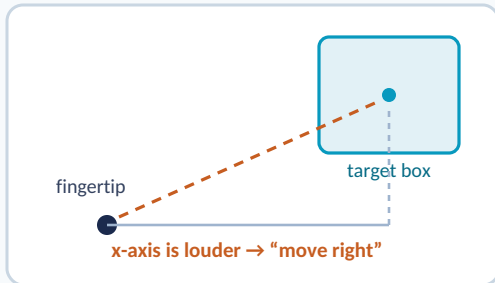
fingertip within 15% of frame on both axes



touching

“Your hand is on the cup. Grasp it”

fingertip inside the box · task completes itself



UNDER THE HOOD

MediaPipe hands wakes up only when the target is near — hand tracking costs 15 ms/frame; across the room it buys nothing

Each frame: a vector from **fingertip to the target's centroid**. Only the dominant axis is spoken — one clear cue, never “right and slightly up”

The guidance **re-plans every frame**: wherever the hand drifts, the next cue corrects from the new position

→ Touch is the system's only autonomous success: the fingertip on the object is the confirmation

Navigation — “Take me to the kitchen”



Goal-directed exploration

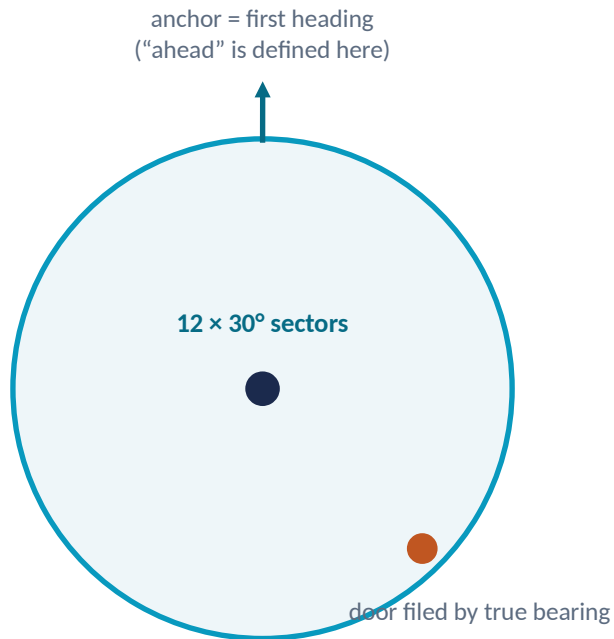
- **No maps. No markers. No SLAM.**
- The user names only the destination; the system explores door-by-door
- Rooms are recognized semantically, from the objects that define them
- Runs at 3 FPS over four models — heavier stack, loaded once and shared

Arrival = ≥ 1 primary or ≥ 2 secondary indicators

Goal	Primary	Secondary
kitchen	fridge, oven, microwave	sink, dining table
bathroom	toilet	sink, toothbrush
bedroom	bed	closet, bedside table
living room	couch, TV	plant, remote

A lone sink never declares a kitchen — but a fridge does. Evidence must be localized: 4 sightings inside one 90° window, so scattered noise can never sum to an arrival.

The compass-anchored 360° room scan



- “Slowly turn to your right, all the way around...” — one guided turn per room
- bearing = heading + in-frame offset → fixed a class of “said left, was right” errors
- Sightings cluster within 30° (across the $\pm 180^\circ$ seam) — one physical door, one announcement
- **Scan ends only on 350° of rotation AND return to start ($\pm 25^\circ$) — compass noise cannot fake completion**
- Blur gate skips fast-pan frames; coaches “slow down” only after 3 in a row

Doors: a custom detector, never trusted alone



Custom door detector

0.946

mAP@50 on 842 held-out images

- 4,213 images: DoorDetect + DeepDoors2 (+827 hard negatives)
- precision 0.968 · recall 0.892
- 2nd model (4-class) = semantic verifier

“Blank walls scored 0.9 as doors” — confidence is not enough

1 • Geometric gates aspect ratio · frame-fill · edge density (smooth → demoted, not dropped)

2 • Semantic verification weak candidates need the verifier to see a door or a handle — kills the lace curtain

3 • Fridge arbitration door-vs-refrigerator claims resolved — no false kitchens

Temporal evidence 2-of-3 frames, ≥3 sightings per scan before a door is ever spoken

The walk: steps, obstacles, and getting through



Distance in steps

- Pinhole model over a 2 m door; focal length from the image's long side (fixed a 33% error)
- “The door is on your right, about 4 steps away” — then a hand cue for the last meter



Obstacle watchdog

- Named objects in the walking lane + SegFormer floor check ($\geq 60\%$ walkable)
- Warnings preempt all door guidance; “step to your right, where it's clear”



Doorway transit

- The camera is the odometer: door gone ~ 2 s + sustained motion = you're through
- Detection flicker alone can never fake progress; then the next room's scan begins

Safety information always precedes the instruction to move.

Speech discipline and asking for help



Two-tier speech policy

- Priority lines always play: warnings, summaries, arrival
- Ambient nudges are dropped, never queued (2.5 s gap · 8 s de-dupe) — audio never lags reality
- Adjacent announcements merge into one utterance — nothing cuts anything off



Multi-door forks: ask the human

“I found two doors: one ahead, and one on your left. Which way is the kitchen?”

- “left” → that door becomes the target
- no answer → nearest straight ahead
- the user is a direction oracle — consulted only when perception is ambiguous

Measured, not hoped: models and voice

Door detector (842 held-out images)

Run	mAP@50	Precision	Recall
8 epochs	0.86	—	—
20 epochs	0.93	—	—
39 epochs	0.946	0.968	0.892

Temporal accumulation over a full scan makes the practical miss rate far lower — and the funnel absorbs the false positives the validation set can't show.

Command parser — 533-utterance corpus

99.4%

exact task type + canonical target

- All 336 noun×prefix and 100 destination×prefix clean commands: 100%
- Mishearings + synonyms + filler: 23/25
- Out-of-scope rejection: 19/20 — and the failures fail safe (re-prompt, not misroute)



Live Demo

What to watch for

silence until spoken to · “find my cup” → grasp · a full room scan · an obstacle interrupting · “we've reached the kitchen”

Limitations and the staged plan they map onto



Known limitations

- Monocular distance is approximate — assumes a 2 m door and a nominal camera FOV
- English-only voice interaction
- Object finding covers a curated COCO vocabulary — not arbitrary items
- Cloud-dependent: all perception and speech run server-side — no connection, no system
- Latency floor: every frame and every utterance crosses the network



Future work

- Arabic voice — Whisper and gTTS already support it; the work is the parser vocabulary
- Vocabulary growth: fine-tune new object and room classes
- On-device inference + offline TTS — removes the cloud dependence
- Faster response — trim the capture-to-speech pipeline for lower delay



Thank You

Questions welcome

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